

DriveBuddyAI Pav-Bhaji Text Classification

Data Analysis and Machine Learning Report

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Project Objective

The objective of this challenge is to build a text-based machine learning classifier that predicts whether an Instagram post is associated with Pav-Bhaji or not. The challenge explicitly requires metadata from the Instagram posts rather than a CNN or pixel-based computer vision model. Therefore, the solution uses textual metadata such as captions and hashtags as the primary predictive information.

1. Dataset Overview

The provided dataset contains Instagram post metadata in **pavbhaji.json** and labeled images organized into two directories. The image directory named **0** represents non-Pav-Bhaji posts, while directory **1** represents Pav-Bhaji posts.

Dataset component	Value
JSON metadata records	1,500
Labeled images	452
Class 0 — Non-Pav-Bhaji	269
Class 1 — Pav-Bhaji	183
Matched image/metadata records used	452

The supplied images were matched to JSON metadata using image filenames extracted from fields such as `display_url`, `thumbnail_src`, and `thumbnail_resources[].src`. Only records for which the supplied image could be matched with metadata were used for supervised classification.

2. Available Metadata Fields

Feature / Field	Description	Use
Caption	Text written for the Instagram post	Primary ML feature
Tags / Hashtags	Keywords associated with the post	Primary ML feature
<code>display_url</code>	URL associated with the post image	Used for matching
<code>thumbnail_src</code>	Thumbnail URL	Used for matching
Likes	Number of likes	EDA / not used
Comments	Number of comments	EDA / not used
Timestamp	Post publication time	EDA / not used
Location	Instagram location metadata	Not used
Owner ID	Instagram account identifier	Not used
Dimensions	Image dimensions	Not used

Feature / Field	Description	Use
is_video	Whether post is a video	Not used

3. Feature Selection and Relevance

The most relevant fields for the classification task are the post caption and associated tags/hashtags because they contain semantic information about the food, dish, ingredients, cuisine, and context described by the Instagram user.

Feature	Relevance	Decision
Caption	Contains descriptive natural-language information about the post	Used
Tags / hashtags	Often identify food names, cuisine, restaurants, locations and related topics.	Used
Likes / comments	Measure engagement rather than the semantic content of the post	Excluded
Owner ID	Identifies an account and may cause account-specific memorization	Excluded
Timestamp	Temporal information has limited direct semantic relevance.	Excluded
Image dimensions	Describe image properties rather than food-related textual meaning	Excluded
Image URL	Used for matching metadata to supplied images, not as a semantic feature	Matching only

4. Target Leakage Prevention

The dataset is related to Pav-Bhaji content and some captions or hashtags directly contain terms such as **pavbhaji**, **pav bhaji**, or variants of the target name. Including these terms directly would make the classification task artificially easy because the model could simply detect the target label in the text rather than learn contextual patterns.

To reduce this target leakage, direct Pav-Bhaji target terms and their common hashtag/hyphen/spacing variants were removed during text cleaning. This encourages the model to learn contextual signals such as Indian food vocabulary, ingredients, dishes, restaurant descriptions and related language.

5. Data Preprocessing

- Loaded Instagram metadata from pavbhaji.json.
- Scanned the supplied image directories and assigned labels based on directory name.
- Matched image filenames with the corresponding JSON metadata.
- Combined caption text and tag/hashtag information into a single textual feature.
- Converted missing textual fields into empty strings.
- Normalized text and removed direct Pav-Bhaji target terms to reduce leakage.
- Used TF-IDF to transform text into numerical features.
- Used both word-level and character-level n-grams to capture semantic phrases and spelling/format variations.

6. Feature Engineering

Word-level TF-IDF: Word n-grams from 1 to 2 words were used. This captures individual terms as well as short phrases such as food-related descriptions and combinations of words.

Character-level TF-IDF: Character n-grams from 3 to 5 characters were used. Character features help capture spelling variations, hashtags, concatenated words, informal Instagram language and partial word patterns.

The two feature representations were combined using sklearn's **FeatureUnion**. The resulting sparse feature representation was passed to the linear classifier.

7. Machine Learning Model

Component	Configuration
Word TF-IDF	1–2 word n-grams
Character TF-IDF	3–5 character n-grams
Minimum document frequency	2
Maximum features	50,000 per vectorizer
TF scaling	Sublinear TF
Classifier	LinearSVC
C	0.1
Class weighting	balanced
Random state	42

Linear Support Vector Classification was selected because it is well suited to high-dimensional sparse TF-IDF representations and is computationally efficient for text classification.

8. Train/Test Split

Split	Total	Class 0	Class 1
Training	361	215	146
Test	91	54	37
Total	452	269	183

A stratified train/test split was used so that the class proportions were approximately preserved between the training and test sets.

9. Cross-Validation Results

Metric	Mean	Standard Deviation
Accuracy	67.02%	4.26%
F1 Score	61.09%	3.95%
ROC-AUC	73.77%	6.20%

Five-fold cross-validation was used to assess how consistently the model performed across different subsets of the labeled data. The results show moderate classification performance with variation between folds, which is expected given the relatively small labeled dataset and noisy social-media text.

10. Held-Out Test Results

Metric	Result
Accuracy	60.44%

Metric	Result
Pav-Bhaji F1 Score	61.70%
ROC-AUC	67.02%

Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	26	28
Actual 1	8	29

The confusion matrix contains 26 true negatives, 28 false positives, 8 false negatives, and 29 true positives. The model therefore identified 29 of the 37 Pav-Bhaji test examples correctly while producing 8 false negative predictions.

11. Interpretation of Learned Features

Inspection of the learned linear SVM coefficients showed that the model uses broader food-related and Indian-cuisine language as contextual signals. Examples of learned positive-side terms included combinations involving food, paneer, Indian food, street food, biryani, food photography and related food vocabulary.

The negative-side features also contained food and Instagram vocabulary, including terms associated with other dishes and regional/social-media contexts. This indicates that the task is challenging because many non-Pav-Bhaji posts are still food-related and may use vocabulary similar to Pav-Bhaji posts.

These feature interpretations should be treated as model correlations, not as proof that any individual word is inherently indicative of Pav-Bhaji.

12. Limitations

- Only 452 supplied images could be used as labeled training examples.
- Instagram captions and hashtags can contain informal language, spelling variations and unrelated promotional text.
- Many non-Pav-Bhaji posts are still food-related, creating semantic overlap between the two classes.
- The dataset may contain collection bias because it is associated with Pav-Bhaji-related Instagram content.
- Metadata-based classification cannot directly inspect the visual appearance of the food.
- Engagement statistics and account identifiers were intentionally excluded from the predictive feature set to reduce the risk of learning dataset-specific or account-specific patterns.
- The held-out test set is relatively small, so the reported test metrics should not be interpreted as universal real-world performance.

13. Final Model and Prediction Output

After evaluation, the selected pipeline was trained on all 452 matched labeled examples using the selected configuration. The trained model was saved as a Joblib file for reuse.

The prediction output contains the image filename, actual label, predicted label and LinearSVC decision score. The decision score provides a continuous measure of the model's distance from its classification boundary and can be used for analysis of prediction confidence.

14. Project Deliverables

File	Purpose
priya_raj.py	Source code for data preparation, feature extraction, training and prediction
priya_raj_data_analysis_report.pdf	Data analysis and machine learning report
priya_raj_drivebuddyai_model.joblib	Trained final classification model
priya_raj_predictions.csv	Predictions and decision scores

15. Conclusion

This project demonstrates a text-based approach for classifying Instagram posts associated with Pav-Bhaji using available post metadata. Caption and hashtag information were selected as the primary predictive features, while potentially noisy or identity-related metadata was excluded from the model.

The final approach combines word-level and character-level TF-IDF features with a LinearSVC classifier. On the held-out test set, the model achieved **60.44% accuracy**, a **61.70% Pav-Bhaji F1 score**, and **67.02% ROC-AUC**. Five-fold cross-validation produced mean accuracy of **67.02%** and mean ROC-AUC of **73.77%**.

The results indicate that Instagram text metadata contains useful signals for the task, while also showing the difficulty caused by noisy and semantically overlapping food-related content. Further improvement could come from additional labeled data, improved text normalization, richer metadata engineering where appropriate, and more systematic model validation.